

Musaad Hydary

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Computer Engineer specializing in full-stack development and systems architecture across blockchain, mobile, and embedded platforms, with expertise building scalable backend APIs and hardware-integrated software systems.

EDUCATION

Queen's University

Kingston, ON

Bachelor of Applied Science in Computer Engineering

September 2021 – April 2026

- Coursework: Operating Systems, Algorithms, Database Management, Distributed Systems, Performance Analysis

SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, Java, HTML, CSS, SQL, MATLAB, Verilog, NIOS II Assembly

Technologies: React, React Native, Node.js, GraphQL, PostgreSQL, Prisma, Redux, TailwindCSS, Firebase, ROS

Tools: Git, Datadog, Mixpanel, Arduino, Roboflow, Vehicle Spy, Expo, Figma, Blender, Intel Quartus Prime, MS Office

ENGINEERING EXPERIENCE

Software Engineer

May 2024 – August 2025

ARENA Exchange

Toronto, ON

- Architected scalable backend APIs with **Node.js**, **GraphQL**, and **Prisma**, integrating **Mixpanel**-powered real-time market alerts that increased user engagement by **24%**
- Built custom profiling tools and implemented **Datadog APM** monitoring to diagnose critical bottlenecks, improving iOS app performance from **49 FPS to 58 FPS (20% increase)** and doubling response times
- Redesigned UI components to enhance mobile app responsiveness and accessibility across the platform

Systems Engineer

September 2023 – April 2025

Queen's AutoDrive Design Team

Kingston, ON

- Engineered lighting components in **C++** with **CAN bus** protocols and optimized data structures, reducing message latency for real-time **ROS** communications to enable safer pedestrian interaction
- Labeled datasets using **Roboflow** to improve object detection accuracy for autonomous vehicle perception
- Led a team of 4 and presented at SAE World Congress 2024 and 2025 on autonomous vehicle safety, securing **2nd place** among 10+ international teams and contributing to industry accessibility standards

PROJECTS

Glucowatch: Hybrid Smartwatch for Diabetes Management

August 2025 – April 2026

- Built a **React Native** app with 12+ custom components implementing **Bluetooth Low Energy** protocols, **HealthKit** integration, and real-time glucose visualization to enable continuous automated health tracking

RISC Processor Synthesis

January 2024 – April 2024

- Designed a complete RISC CPU with bidirectional bus, ALU, control circuitry, and I/O ports using **Intel Quartus Prime**, **Verilog** and **ModelSim**, demonstrating mastery of computer architecture fundamentals

RapidCraft: AI-Powered Resume Generator

January 2024

- Built a full-stack resume generator with **React** and **Node.js** using **OpenAI's API** to dynamically generate professionally-formatted **LaTeX** resumes in seconds, making resume creation accessible to anyone

Bookkeeper: Receipt Classifier & Budget Optimizer

January 2023

- Architected a financial platform with **Python Optical Character Recognition**, **React**, **Redux**, and **Firebase** to automatically classify receipts and generate budget recommendations, eliminating manual expense tracking

EXTRACURRICULAR LEADERSHIP

Co-President

July 2025 – June 2026

Queen's Music Production Club (QWave)

Kingston, ON

- Led 20+ studio jam sessions and production workshops, grew membership from 150 to 250+ students, and managed DAW software distribution for the club community

Public Speaker

September 2024 – August 2025

Toastmasters International at High Park Speakers

Toronto, ON

- Won **1st place** in a speech competition among 12+ Toronto clubs and earned **Triple Crown award** for completing communication and leadership tracks while mentoring club members